Chapter 10 · Trigonometric identities and equations

Exact values

Derive the sine, cosine and tangent of 30°, 45° and 60°; use the unit circle to extend these ideas to other angles.

Trigonometric identities

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$
$$\sin^2\theta + \cos^2\theta = 1$$

Trigonometric equations

Solve equations such as $\sin(n\theta) = k$ and $\sin(\theta \pm \alpha) = k$ over a specified interval.

Quadratic equations

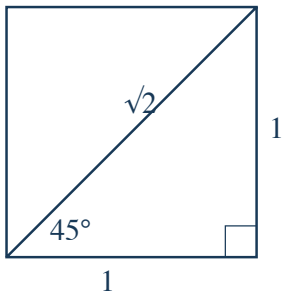
Solve equations that are quadratic in sine, cosine or tangent. Check the range of the function and the interval for the angle.

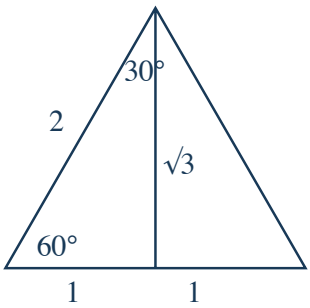
sin/cos/tan of 30°, 45°, 60° — Luisa · 3

You will frequently encounter 30˚, 60˚ and 45˚ in geometric problems.
Why?

Why?

Although you have a calculator, you need to know how to derive these values.
Draw half a unit square and half an equilateral triangle of side 2.



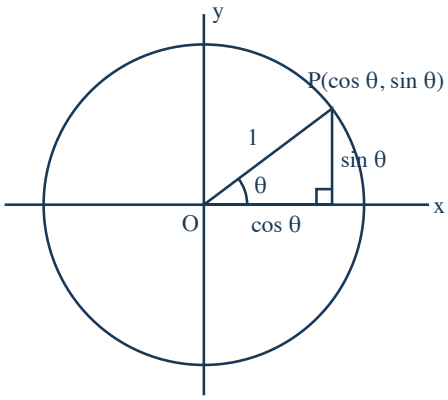


$\sin 45° =$		$\cos 45° =$		$\tan 45° =$	
$\sin 30° =$		$\cos 30° =$		$\tan 30° =$	
$\sin 60° =$		$\cos 60° =$		$\tan 60° =$	

The Unit Circle and Trigonometry — Luisa · 4

The unit circle

A point P on the circle of radius 1 has coordinates $(\cos\theta, \sin\theta)$. Positive angles turn anticlockwise from the positive x-axis.



For an acute angle

$$\cos\theta = \frac{\text{adjacent}}{1}, \qquad \sin\theta = \frac{\text{opposite}}{1}$$

What is the gradient of OP?

Gradient

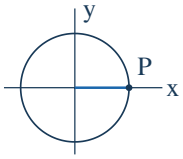
The coordinates extend sine and cosine to angles in every quadrant. Tangent is undefined when cosine is zero.

Mini-Exercise **Luisa · 5**

Use the unit circle to determine each value, using 0, +ve, −ve, 1, −1 or undefined.

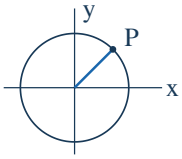
Recall: $P = (\cos\theta, \sin\theta)$; $OP = 1$, and the gradient of OP is $\tan\theta$.
Cosine gives the x-value; sine gives the y-value.

1. $\theta = 0°$



$\cos\theta = 1$
$\sin\theta = 0$
$\tan\theta = 0$

2. $0° < \theta < 90°$



2. values

3. $\theta = 90°$ | 3. diagram | 3. values

4. $90° < \theta < 180°$ | 4. diagram | 4. values

5. $\theta = 180°$ | 5. diagram | 5. values

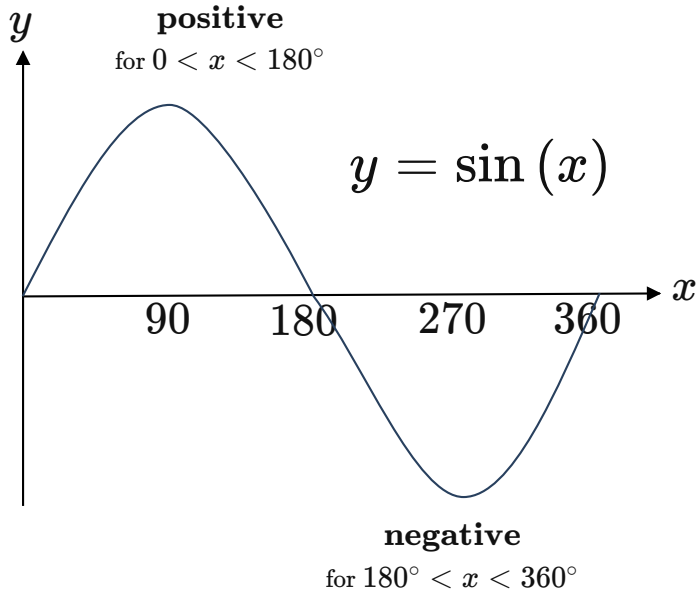
6. $180° < \theta < 270°$ | 6. diagram | 6. values

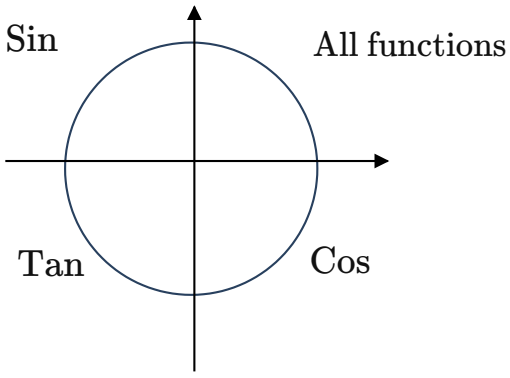
7. $\theta = 270°$ | 7. diagram | 7. values

8. $270° < \theta < 360°$ | 8. diagram | 8. values

The Unit Circle and Trigonometry **Luisa · 6**

The unit circles explains the behaviour of these trigonometric graphs beyond $90°$.
However, the easiest way to remember whether $\sin(x)$, $\cos(x)$, $\tan(x)$ are positive or negative is to just do a **very quick sketch (preferably mentally!)** of the corresponding graph.





Note: The textbook uses something called '*CAST diagrams*', which is an acronym for which quadrants the **C**osine, **A**ll, **S**ine and **T**angent are positive

A Few Trigonometric Angle Laws

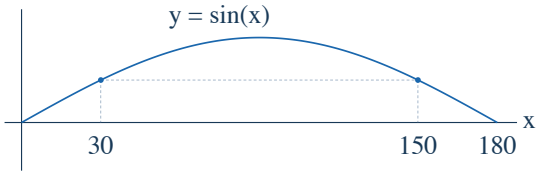
Luisa · 7

These laws follow from a quick sketch of a trigonometric graph. They are a convenience so you need not draw a graph every time. Memorising them helps you answer exam questions faster.

1. $\sin x = \sin(180° - x)$.

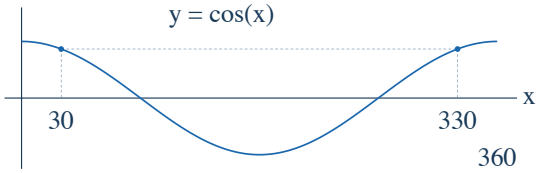
For example, $\sin 150° = \sin 30°$.

We saw this in the previous chapter: the ambiguous case when using the sine rule.



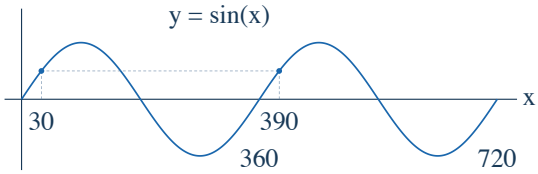
2. $\cos x = \cos(360° - x)$.

For example, $\cos 330° = \cos 30°$.



3. Sine and cosine repeat every 360°; tangent repeats every 180°.

For example, $\sin 390° = \sin 30°$.



4. $\sin x = \cos(90° - x)$.

For example, $\sin 50° = \cos 40°$.

Recall from the previous chapter: cosine is the sine of the complementary angle. This was not covered in the textbook, but appeared unexpectedly in a C3 exam.

Examples

Luisa · 8

Without a calculator, find each exact value.

(a) $\tan 225°$

a

(b) $\tan 210°$

b

(c) $\sin 150°$

c

(d) $\cos 300°$

d

(e) $\sin(-45°)$

e

(f) $\cos 750°$

f

(g) $\cos 120°$

g

Sine and cosine repeat every 360°; tangent repeats every 180°. Use a reference angle and the sign in the quadrant.

Sine and tangent are odd functions: $f(-x) = -f(x)$. Cosine is even: $\cos(-x) = \cos x$.

Without a calculator, work out the value of each below.

a. $\cos 315°$ — a

b. $\sin 420°$ — b

c. $\tan(-120°)$ — c

d. $\tan(-45°)$ — d

e. $\sin 135°$ — e

$\sin(x) = \sin(180 - x)$

$\cos(x) = \cos(360 - x)$

sin, cos repeat every 360° but tan every 180°

Exercise 10B – Y1 Page 209 **Luisa · 10**

Use a reference angle of 30°, 45° or 60° to find each exact value.

(a) $\sin 135°$ a

(b) $\sin(-60°)$ b

(c) $\sin 330°$ c

(d) $\sin 420°$ d

(e) $\sin(-300°)$ e

(f) $\cos 120°$ f

(g) $\cos 300°$ g

(h) $\cos 225°$ h

(i) $\cos(-210°)$ i

(j) $\cos 495°$ j

(k) $\tan 135°$ k

(l) $\tan(-225°)$ l

(m) $\tan 210°$ m

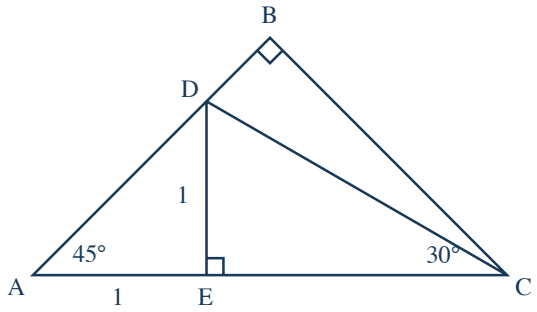
(n) $\tan 300°$ n

(o) $\tan(-120°)$ o

Exercise 10A/B – Y1 Page 207, 209 **Luisa · 11**

Challenge: exact values for 15°

ABC is an isosceles right-angled triangle, with the right angle at B. D lies on AB and E lies on AC. AE = DE = 1, DE is perpendicular to AC, and angle ACD = 30°.



(a)

Find the exact lengths CE, DC, BC and DB.

a

(b)

Find angle BCD.

b

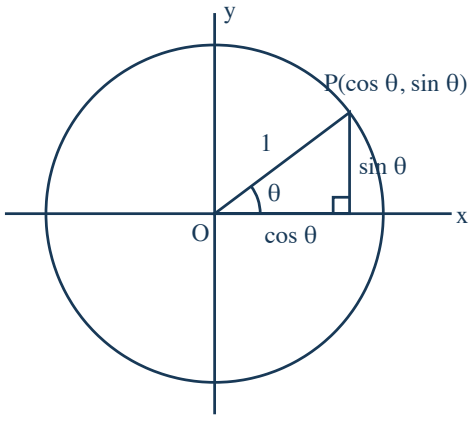
(c)

Find exact values of $\sin 15^\circ$ and $\cos 15^\circ$.

c

Trigonometric Identities **Luisa · 12**

Two trigonometric identities



Here $\sin^2\theta$ means $(\sin\theta)^2$, not $\sin(\theta^2)$.

1.

Opposite divided by adjacent

1

This identity holds whenever cosine is nonzero.

2.

Pythagoras on the unit circle

2

The identity holds for every real angle.

Application of identities #1: Proofs

Luisa · 13

Prove the identity (where the expressions are defined).

$1 - \tan\theta \sin\theta \cos\theta \equiv \cos^2\theta$

Method

Start with one side and transform it into the other.
Replace tangent by sine divided by cosine.

$\tan\theta = \dfrac{\sin\theta}{\cos\theta}$

Proof

The symbol $\equiv$ denotes an identity: equality for every angle in the common domain. Here cosine must be non-zero because tangent appears.

More Examples

Luisa · 14

Prove the identity where both sides are defined.

$\tan\theta + \dfrac{1}{\tan\theta} \equiv \dfrac{1}{\sin\theta \cos\theta}$

Simplify.

$5 - 5\sin^2\theta$

Look for $1 - \sin^2\theta = \cos^2\theta$ or $1 - \cos^2\theta = \sin^2\theta$.

Combine fractions using a common denominator.

Test Your Understanding — Luisa · 15

Prove each identity where its expressions are defined.

$\tan^2\theta \equiv \dfrac{1}{\cos^2\theta} - 1$

For $0 < x < 90°$, prove:

$\dfrac{\tan x \cos x}{\sqrt{1-\cos^2 x}} = 1$

$\dfrac{\cos^4\theta - \sin^4\theta}{\cos^2\theta} \equiv 1 - \tan^2\theta$

Exercise 10C – Y1 Page 211-212 — Luisa · 16

10.

Eliminate θ to give an equation relating x and y in each case.

(a)

$x = \sin\theta, \quad y = \cos\theta$

a

(b)

$x = \sin\theta, \quad y = 2\cos\theta$

b

(c)

$x = \sin\theta, \quad y = \cos^2\theta$

c

(d)

$x = \sin\theta, \quad y = \tan\theta$

d

(e)

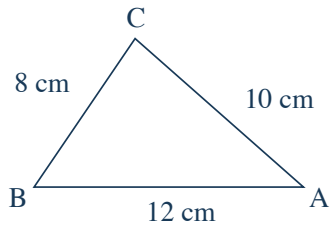
$x = \sin\theta + \cos\theta$

$y = \cos\theta - \sin\theta$

e

Exercise 10C – Y1 Page 211-212 **Luisa · 17**

11. Triangle ABC has $AB = 12\,\text{cm}$, $BC = 8\,\text{cm}$ and $AC = 10\,\text{cm}$.



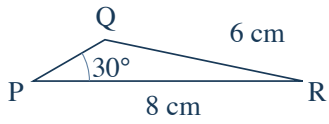
a. Show that $\cos B = \frac{9}{16}$. (3 marks)

11a.

b. Hence find the exact value of $\sin B$. (2 marks)

11b.

12. Triangle PQR has $PR = 8\,\text{cm}$, $QR = 6\,\text{cm}$ and $\angle QPR = 30°$.



a. Show that $\sin Q = \frac{2}{3}$. (3 marks)

12a.

b. Given that Q is obtuse, find the exact value of $\cos Q$. (2 marks)

12b.

Extension **Luisa · 18**

[MAT 2008 1C] For which angles in $0 \leq \theta < 2\pi$ are these simultaneous equations solvable?

$x\cos\theta - y\sin\theta = 2$

$x\sin\theta + y\cos\theta = 1$

A: all angles; B: all except one; C: all except two; D: all except three.

Answer

No division by a potentially zero sine or cosine is required. Dividing by one could incorrectly exclude valid angles.

Solving Trigonometric Equations — Luisa · 19

Solve $\sin\theta = \frac{1}{2}$ for $0 \leq \theta \leq 360°$.

An inverse trigonometric function on a calculator returns only the principal value.

$\sin\theta = \sin(180° - \theta)$

$\cos\theta = \cos(360° - \theta)$

Sine and cosine repeat every 360°; tangent repeats every 180°.

Solve $5\tan\theta = 10$ for $-180° \leq \theta < 180°$.

Tangent repeats every 180°. Always check the stated interval.

Slightly Harder Ones... — Luisa · 20

Solve $\sin\theta = -\frac{1}{2}$ for $0 \leq \theta \leq 360°$.

The principal value -30° is outside the interval. The required answers are 210° and 330°.

Solve $\sin\theta = \sqrt{3}\cos\theta$ for $0 \leq \theta \leq 360°$.

Express the equation in one trigonometric function.

Test Your Understanding — Luisa · 21

Solve $2\cos\theta = \sqrt{3}$ for $0 \leq \theta \leq 360°$.

Solve $\sqrt{3}\sin\theta = \cos\theta$ for $-180° \leq \theta \leq 180°$.

Exercise 10D – Y1 215-216 — Luisa · 22

7.

Solve $2\cos x = 3\sin x$ for $-180^\circ \leq x \leq 180^\circ$.

Student A obtains $\tan x = 3/2$ and $x = 56.3^\circ$ or -123.7°.

Student B squares both sides and obtains $\sin x = \pm\sqrt{(4/13)}$, giving $x = \pm 33.7^\circ$ or $\pm 146.3^\circ$.

(a)
Identify Student A's mistake.

7(a)

(b)
Identify Student B's mistake and its effect.

7(b)

(c)
Give the correct answers.

7(c)

9.

Solve $\tan^2\theta = 9$ for $0 < \theta < 360°$, to 1 decimal place.

9

Exercise 10D – Y1 215-216

10(a)

Show that $4\sin^2 x - 3\cos^2 x = 2$ becomes $7\sin^2 x = 5$.

10(a)

10(b)

Hence solve for $0 \leq x \leq 360^\circ$, giving answers to 1 decimal place.

10(b)

11(a)

Show that $2\sin^2 x + 5\cos^2 x = 1$ becomes $3\sin^2 x = 4$.

11(a)

11(b)

Explain why there are no real solutions.

11(b)

Harder Equations

Harder questions replace the angle θ with a linear expression.

Solve $\cos 3x = -\frac{1}{2}$ in the interval $0 \leq x \leq 360°$.

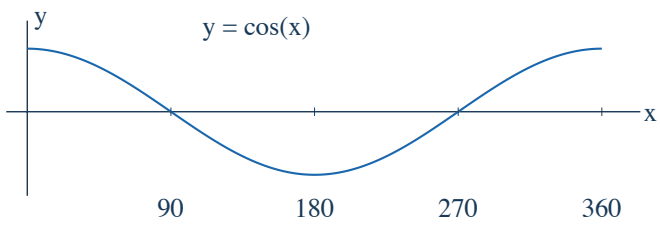
STEP 1. Adjust the range for the expression inside the cosine.

STEP 2. Immediately after applying the inverse trigonometric function, find every solution up to the end of the interval—before dividing by 3.

STEP 3. Apply the final manipulation to every value.

1.

In general, sine, cosine and tangent each give a pair of values per 360°, except when $\cos\theta = \pm 1$ or $\sin\theta = \pm 1$.
Find the first pair using $\sin(180^\circ - \theta)$ or $\cos(360^\circ - \theta)$, then keep adding 360° to generate new pairs.



Further Examples

Luisa · 25

Solve $\sin(2x + 30°) = \frac{1}{\sqrt{2}}$ for $0 \leq x \leq 360°$.

Solve $\sin x = 2\cos x$ for $0 \leq x < 300°$.

Transform both endpoints of the interval before finding all the angles.

Test Your Understanding

Luisa · 26

Solve $\cos(3x - 10°) = -0.4$ for $0 \leq x < 180°$. Show each step and give answers to 1 decimal place. (7 marks)

Adjust the interval

Find every intermediate angle

Add 10° and divide each angle by 3

4.

Solve for $0 \leq x \leq 180^\circ$.

(a)

$\sin(x + 20^\circ) = \frac{1}{2}$

4(a)

(b)

Give answers to 1 decimal place.

$\cos 2x = -0.8$

4(b)

5(a)

Sketch $y = \sin(x + 60^\circ)$ for $0 \leq x \leq 360^\circ$.

5(a)

5(b)

Give the exact axis intercepts.

5(b)

5(c)

Solve $\sin(x + 60^\circ) = 0.55$ on the same interval, to 1 decimal place.

5(c)

6(a)

Given $4 \sin x = 3 \cos x$, find $\tan x$.

6(a)

6(b)

Solve $4 \sin 2\theta = 3 \cos 2\theta$ for $0 \leq \theta \leq 360^\circ$, to 1 decimal place.

6(b)

7

The equation $\tan(kx) = -1/\sqrt{3}$, where $k > 0$, has a solution at $x = 60^\circ$.

(a)

Find a possible k.

7(a)

(b)

Is this the only possible value? Justify your answer.

7(b)

The period of tan(kx) is $180^\circ/k$, so it decreases as k increases.

Quadratics in sin/cos/tan

Luisa · 29

Solve $5\sin^2 x + 3\sin x - 2 = 0$ for $0 \leq x \leq 360°$.

Method 1: substitute

Method 2: factorise directly

"Quadratic in sine" means that replacing $\sin x$ with a single variable produces a quadratic. Similarly $x - 2\sqrt{x} + 1 = 0$ is quadratic in $\sqrt{x}$.

More Examples

Luisa · 30

Solve $\tan^2 \theta = 4$ for $0 \leq \theta \leq 360°$.

Include both signs of the square root. Tangent repeats every 180°; discard angles outside the stated interval.

Solve $2\cos^2 x + \sin x = 3\sin^2 x$ for $-180° \leq x \leq 180°$.

2

Use the identity to express the equation in one trigonometric function, then use the quadratic formula.

Test Your Understanding — Luisa · 31

a. Show that $5\sin x = 1 + 2\cos^2 x$ can be written as $2\sin^2 x + 5\sin x - 3 = 0$.

a

b. Solve $2\sin^2 x + 5\sin x - 3 = 0$ for $0 \leq x < 360°$.

b

Exercise 10F – Y1 P221-222 — Luisa · 32

5.

Find all solutions of $8\sin^2 x + 6\cos x - 9 = 0$ for $0 \leq x \leq 360°$, to 1 decimal place. (6 marks)

5: rearrange

5: solve

6.

Find all solutions of $\sin^2 x + 1 = \frac{7}{2}\cos^2 x$ for $0 \leq x \leq 360°$, to 1 decimal place.

6: rearrange

6: solve

7.

Show that $2\cos^2 x + \cos x - 6 = 0$ has no solutions.

7

8(a)

Show that $\cos^2 x = 2 - \sin x$ can be written as $\sin^2 x - \sin x + 1 = 0$. (2 marks)

8(a)

8(b)

Hence show that the original equation has no solutions. (3 marks)

8(b)

9.

$\tan^2 x - 2\tan x - 4 = 0$

(a)

Show that $\tan x = p \pm \sqrt{q}$, finding p and q.

a

(b)

Hence solve the equation for $0 \leq x \leq 540^\circ$.

b: angles

b: solutions

1.

[MAT 2010 1C]

How many solutions are there for $0 \leq x < 360°$?

$\sin^2 x + 3 \sin x \cos x + 2 \cos^2 x = 0$

1

2.

[MAT 2014 1E]

Find the largest value, for real x, of

$(4 \sin^2 x + 4 \cos x + 1)^2$

2